



Qualification Pack



AI Ethics Specialist

QP Code: MSU/ITS/Q2001

Version: 1.0

NSQF Level: 6

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MSU/ITS/Q2001: AI Ethics Specialist

Brief Job Description

The qualification is intended for individuals aspiring to work as AI ethics specialists or those engaged in AI system development and deployment. It covers stakeholder engagement, collaboration with technical experts, ethical risk assessment, the application of AI ethics frameworks, and conducting ethical impact assessments. This qualification equips professionals, including AI developers, data scientists, and interdisciplinary groups, with the practical skills to navigate ethical considerations in AI projects. It fosters ethical awareness, communication, and responsible AI practices in a rapidly evolving landscape.

Personal Attributes

To excel as an AI Ethics Specialist, one must possess a strong sense of ethical integrity, critical thinking, and effective communication skills to navigate complex ethical considerations in AI. A collaborative mindset, adaptability, and empathy are crucial for engaging with diverse stakeholders and ensuring responsible AI practices. The ability to assess ethical risks and apply AI ethics frameworks with transparency and foresight is essential in this rapidly evolving field.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MSU/ITS/N2003: Identify and Assess the Ethical Risks of AI Systems](#)
2. [MSU/ITS/N2002: Develop and Implement Ethical Guidelines and Policies for AI Development and Use](#)
3. [MSU/ITS/N1001: Facilitate Ethical Communication and Educational Outreach with Stakeholders](#)
4. [MSU/ITS/N2001: Collaborate with Cross-Functional Teams for Ethical AI Development](#)
5. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	ARTIFICIAL INTELLIGENCE
Occupation	Artificial Intelligence
Country	India



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NSQF Level	6
Credits	24
Aligned to NCO/ISCO/ISIC Code	NCO-2015/1330.9900
Minimum Educational Qualification & Experience	Completed 3 year UG degree (with AI/ML/Data Science) OR Completed 4 year UG program (with AI/ML/Data Science) OR Pursuing 1st year of PG- Eng (Pursuing 1st year of PG degree / PG Diploma in AI/ML/Data Science (after 3- year UG degree) OR Completed 3 year UG degree (with Knowledge of IT and Programming) with 2 Years of experience AI/ML/Data Science OR Previous relevant Qualification of NSQF Level (5.5) (in AI/ML/Data Science) with 1-2 Years of experience in AI/ML/Data Science OR Previous relevant Qualification of NSQF Level (6) (in the field of IT/ITeS) with 2 Years of experience in AI/ML/Data Science
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	31/10/2028
NSQF Approval Date	01/11/2023
Version	1.0
Reference code on NQR	QG-06-IT-01274-2023-V1-MSU
NQR Version	1



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MSU/ITS/N2003: Identify and Assess the Ethical Risks of AI Systems

Description

This OS unit is about engaging stakeholder to obtain ethical insights, assess ethical risks, analyse ethical impact and to recommend mitigation strategies.

Scope

The scope covers the following :

- Engage Stakeholders and Obtain Ethical Insights
- Assess Ethical Risks
- Analyse Ethical Impact and Recommend Mitigation Strategies

Elements and Performance Criteria

Engage Stakeholders and Obtain Ethical Insights

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and map key stakeholders in AI development, deployment, and regulation, illustrating their roles, ethical concerns, and perspective
- PC2.** Collaborate with various stakeholders, including AI developers, legal teams, and technical experts, to gain insights into the technical aspects of AI systems and identify potential ethical risks
- PC3.** Forge robust relationships with senior executives and CxOs, leveraging Data Science background to connect on technical and strategic levels
- PC4.** Collaborate with the interdisciplinary team to identify synergies between Data Science and Responsible AI for client strategies
- PC5.** Conduct thorough consultations to comprehend client needs, leveraging Data Science acumen to decipher complex requirements
- PC6.** Provide expert guidance on embedding ethical considerations in AI development and governance strategies
- PC7.** Deliver tailor-made solutions that integrate Data Science and AI ethics, aligning them with client's unique objectives

Assess Ethical Risks

To be competent, the user/individual on the job must be able to:

- PC8.** Research and analyse the potential ethical risks associated with AI systems
- PC9.** Assess the data privacy and security aspects of AI systems to ensure that they respect privacy rights, protect data, and have robust security measures in place
- PC10.** Examine the data used to train AI algorithms and evaluate potential biases and fairness issues that may arise in AI systems
- PC11.** Evaluate the social and societal impacts of AI systems by assessing the potential consequences for employment, inequality, human rights, and environmental sustainability



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- PC12.** Review existing AI ethics guidelines and frameworks to understand the normative implications and practical guidance available for addressing ethical issues in AI development and use
- PC13.** Utilise hands-on experience using techniques such as data bias testing (e.g., for under-represented groups, proxy variables, recall bias, skew etc), utility (e.g., SHAP values, LIME)

Analyse Ethical Impact and Recommend Mitigation Strategies

To be competent, the user/individual on the job must be able to:

- PC14.** Conduct ethical impact assessments of AI systems to uncover potential ethical challenges
- PC15.** Prioritise ethical risks considering their potential impact on human rights, societal values, fairness, and environmental considerations
- PC16.** Analyse real-world case studies to comprehend and evaluate ethical challenges in AI applications
- PC17.** Provide recommendations and guidance to the organisation based on their assessments, suggesting measures to mitigate ethical risks, ensuring compliance with ethical principles, and promoting responsible AI development and use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Stakeholder Identification and Analysis
- KU2.** Technical Insights into AI Systems
- KU3.** Executive-Level Engagement and Communication
- KU4.** Interdisciplinary Collaboration for Client Strategies
- KU5.** Client Needs Comprehension and Translation
- KU6.** AI Ethics Frameworks and Principles
- KU7.** Customized Ethical Solution Design
- KU8.** Ethical Risk Assessment in AI
- KU9.** Data Privacy and Security Compliance
- KU10.** Bias Detection and Fairness Evaluation
- KU11.** Societal Impact Assessment in AI
- KU12.** AI Ethics Guidelines Familiarity
- KU13.** Hands-On Data Bias Testing and Utilisability Techniques
- KU14.** Ethical Impact Assessment for AI Systems
- KU15.** Ethical Risk Prioritization
- KU16.** Real-World Case Study Analysis
- KU17.** Ethical Recommendations and Mitigation Strategies



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Engage Stakeholders and Obtain Ethical Insights</i>	10	10	2	5
PC1. Identify and map key stakeholders in AI development, deployment, and regulation, illustrating their roles, ethical concerns, and perspective	-	-	-	-
PC2. Collaborate with various stakeholders, including AI developers, legal teams, and technical experts, to gain insights into the technical aspects of AI systems and identify potential ethical risks	-	-	-	-
PC3. Forge robust relationships with senior executives and CxOs, leveraging Data Science background to connect on technical and strategic levels	-	-	-	-
PC4. Collaborate with the interdisciplinary team to identify synergies between Data Science and Responsible AI for client strategies	-	-	-	-
PC5. Conduct thorough consultations to comprehend client needs, leveraging Data Science acumen to decipher complex requirements	-	-	-	-
PC6. Provide expert guidance on embedding ethical considerations in AI development and governance strategies	-	-	-	-
PC7. Deliver tailor-made solutions that integrate Data Science and AI ethics, aligning them with client's unique objectives	-	-	-	-
<i>Assess Ethical Risks</i>	10	15	5	10
PC8. Research and analyse the potential ethical risks associated with AI systems	-	-	-	-
PC9. Assess the data privacy and security aspects of AI systems to ensure that they respect privacy rights, protect data, and have robust security measures in place	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Examine the data used to train AI algorithms and evaluate potential biases and fairness issues that may arise in AI systems	-	-	-	-
PC11. Evaluate the social and societal impacts of AI systems by assessing the potential consequences for employment, inequality, human rights, and environmental sustainability	-	-	-	-
PC12. Review existing AI ethics guidelines and frameworks to understand the normative implications and practical guidance available for addressing ethical issues in AI development and use	-	-	-	-
PC13. Utilise hands-on experience using techniques such as data bias testing (e.g., for under-represented groups, proxy variables, recall bias, skew etc), utility (e.g., SHAP values, LIME)	-	-	-	-
<i>Analyse Ethical Impact and Recommend Mitigation Strategies</i>	10	15	3	5
PC14. Conduct ethical impact assessments of AI systems to uncover potential ethical challenges	-	-	-	-
PC15. Prioritise ethical risks considering their potential impact on human rights, societal values, fairness, and environmental considerations	-	-	-	-
PC16. Analyse real-world case studies to comprehend and evaluate ethical challenges in AI applications	-	-	-	-
PC17. Provide recommendations and guidance to the organisation based on their assessments, suggesting measures to mitigate ethical risks, ensuring compliance with ethical principles, and promoting responsible AI development and use	-	-	-	-
NOS Total	30	40	10	20



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National Occupational Standards (NOS) Parameters

NOS Code	MSU/ITS/N2003
NOS Name	Identify and Assess the Ethical Risks of AI Systems
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2028
NSQC Clearance Date	01/11/2023



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MSU/ITS/N2002: Develop and Implement Ethical Guidelines and Policies for AI Development and Use

Description

This OS unit is about staying updated in AI ethics, crafting and communicating ethical standards and guidelines and ensuring ethical integration and compliance in AI Development.

Scope

The scope covers the following :

- Stay Updated in AI Ethics
- Craft and Communicate Ethical Standards and Guidelines for AI
- Ensure Ethical Integration and Compliance in AI Development

Elements and Performance Criteria

Stay Updated in AI Ethics

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor and analyse the latest developments in AI ethics, encompassing emerging trends and controversies
- PC2.** Engage with AI ethics forums, conferences, and online communities to exchange insights and perspectives
- PC3.** Participate in continuous learning through webinars, podcasts, and reading materials on AI ethics
- PC4.** Collaboratively map out the ethical implications of evolving AI technologies with fellow professionals

Craft and Communicate Ethical Standards and Guidelines for AI

To be competent, the user/individual on the job must be able to:

- PC5.** Define and communicate ethical standards for responsible AI development, including fairness and transparency
- PC6.** Formulate clear and comprehensive ethical guidelines for responsible AI development and deployment
- PC7.** Align ethical guidelines with international standards, legal frameworks, cultural values, and social, political, and environmental impact
- PC8.** Ensure adherence to ethical principles in AI projects and applications
- PC9.** Communicate ethical guidelines effectively to stakeholders involved in AI projects

Ensure Ethical Integration and Compliance in AI Development

To be competent, the user/individual on the job must be able to:

- PC10.** Collaborate with technical teams to ensure ethical considerations are seamlessly integrated into Data Science and AI projects
- PC11.** Design and implement AI systems with a strong emphasis on transparency, fairness, accountability, and privacy, utilizing your Data Science expertise



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- PC12.** Utilize your expertise in supervised learning techniques, such as regression, classification, and econometric models, to address ethical challenges and considerations within these frameworks
- PC13.** Apply your proficiency in unsupervised learning clustering techniques to identify and address potential ethical issues that may arise within unstructured data. This skill will be beneficial when collaborating with multidisciplinary teams to gather diverse perspectives and insights on AI ethics
- PC14.** Ensure adherence to ethical principles in AI projects and applications
- PC15.** Integrate ethical principles at every stage of the AI project lifecycle
- PC16.** Monitor compliance with ethical standards and adapt them based on feedback
- PC17.** Collaborate with cross-functional teams to create industry-relevant, responsible AI frameworks
- PC18.** Advocate for the inclusion of diverse perspectives and expertise in AI development
- PC19.** Develop and execute implementation plans to ensure ethical policies are effectively applied
- PC20.** Collaborate with legal experts to ensure alignment with regulations and legal compliance
- PC21.** Analyse case studies to understand the influence of ethics on shaping AI policy and governance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Foundational Understanding of AI Ethics
- KU2.** Peer Engagement and Validation in AI Ethics
- KU3.** Monitoring and Analysis of AI Ethics Developments
- KU4.** Engagement with AI Ethics Communities
- KU5.** Continuous Learning in AI Ethics
- KU6.** Mapping Ethical Implications in Evolving AI Technologies
- KU7.** Ethical Standards Definition for AI Development
- KU8.** Ethical Guidelines Formulation
- KU9.** Alignment of Ethics with International Standards
- KU10.** Ethical Adherence in AI Projects
- KU11.** Effective Ethical Guideline Communication
- KU12.** Ethical Integration in AI Development
- KU13.** Transparency, Fairness, Accountability, and Privacy in AI Systems
- KU14.** Supervised Learning for Ethical Challenges
- KU15.** Unsupervised Learning for Ethical Issues in Unstructured Data
- KU16.** Ethical Integration in AI Lifecycle
- KU17.** Ethical Compliance Monitoring and Adaptation
- KU18.** Cross-Functional Collaboration for Responsible AI Frameworks
- KU19.** Diversity and Inclusion Advocacy in AI Development
- KU20.** Implementation of Ethical Policies



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KU21. Alignment with Legal and Regulatory Compliance

KU22. Case Study Analysis on Ethics in AI Policy and Governance



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Stay Updated in AI Ethics</i>	7	10	2	5
PC1. Monitor and analyse the latest developments in AI ethics, encompassing emerging trends and controversies	-	-	-	-
PC2. Engage with AI ethics forums, conferences, and online communities to exchange insights and perspectives	-	-	-	-
PC3. Participate in continuous learning through webinars, podcasts, and reading materials on AI ethics	-	-	-	-
PC4. Collaboratively map out the ethical implications of evolving AI technologies with fellow professionals	-	-	-	-
<i>Craft and Communicate Ethical Standards and Guidelines for AI</i>	7	10	3	5
PC5. Define and communicate ethical standards for responsible AI development, including fairness and transparency	-	-	-	-
PC6. Formulate clear and comprehensive ethical guidelines for responsible AI development and deployment	-	-	-	-
PC7. Align ethical guidelines with international standards, legal frameworks, cultural values, and social, political, and environmental impact	-	-	-	-
PC8. Ensure adherence to ethical principles in AI projects and applications	-	-	-	-
PC9. Communicate ethical guidelines effectively to stakeholders involved in AI projects	-	-	-	-
<i>Ensure Ethical Integration and Compliance in AI Development</i>	16	20	5	10
PC10. Collaborate with technical teams to ensure ethical considerations are seamlessly integrated into Data Science and AI projects	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Design and implement AI systems with a strong emphasis on transparency, fairness, accountability, and privacy, utilizing your Data Science expertise	-	-	-	-
PC12. Utilize your expertise in supervised learning techniques, such as regression, classification, and econometric models, to address ethical challenges and considerations within these frameworks	-	-	-	-
PC13. Apply your proficiency in unsupervised learning clustering techniques to identify and address potential ethical issues that may arise within unstructured data. This skill will be beneficial when collaborating with multidisciplinary teams to gather diverse perspectives and insights on AI ethics	-	-	-	-
PC14. Ensure adherence to ethical principles in AI projects and applications	-	-	-	-
PC15. Integrate ethical principles at every stage of the AI project lifecycle	-	-	-	-
PC16. Monitor compliance with ethical standards and adapt them based on feedback	-	-	-	-
PC17. Collaborate with cross-functional teams to create industry-relevant, responsible AI frameworks	-	-	-	-
PC18. Advocate for the inclusion of diverse perspectives and expertise in AI development	-	-	-	-
PC19. Develop and execute implementation plans to ensure ethical policies are effectively applied	-	-	-	-
PC20. Collaborate with legal experts to ensure alignment with regulations and legal compliance	-	-	-	-
PC21. Analyse case studies to understand the influence of ethics on shaping AI policy and governance	-	-	-	-
NOS Total	30	40	10	20



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National Occupational Standards (NOS) Parameters

NOS Code	MSU/ITS/N2002
NOS Name	Develop and Implement Ethical Guidelines and Policies for AI Development and Use
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2028
NSQC Clearance Date	01/11/2023



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MSU/ITS/N1001: Facilitate Ethical Communication and Educational Outreach with Stakeholders

Description

This OS unit is about providing educational outreach and ethical guidance.

Scope

The scope covers the following :

- Provide Educational Outreach and Ethical Guidance

Elements and Performance Criteria

Provide Educational Outreach and Ethical Guidance

To be competent, the user/individual on the job must be able to:

- PC1.** Utilise visual aids, storytelling, and practical examples to enhance the understanding of AI ethical concepts
- PC2.** Address ethical concerns transparently and foster open dialogue with stakeholders
- PC3.** Educate team members about ethical AI practises, fostering a culture of responsibility and ethical consciousness within the organisation
- PC4.** Communicate the importance of ethical principles in AI projects and applications
- PC5.** Develop training and educational materials on AI ethics for internal and external audiences
- PC6.** Develop thought-provoking content, such as articles and presentations, that bridges the gap between Data Science and Responsible AI
- PC7.** Conduct training sessions and programs to raise awareness among AI developers and users about ethical considerations
- PC8.** Mentor junior consultants, drawing from Data Science experience to nurture their growth
- PC9.** Stay informed about the latest research and guidelines in AI ethics to align policies with evolving standards
- PC10.** Provide timely advice on the ethical implications of AI decisions to guide decision-makers
- PC11.** Lead cross-functional teams, comprising Data Scientists, AI engineers, designers, and other experts, to cultivate a culture of ethical awareness, guidance, and responsible AI practices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Stakeholder Engagement Strategies in AI Ethics
- KU2.** Visual Aids and Storytelling in AI Ethics Communication
- KU3.** Transparency and Open Dialogue on Ethical Concerns
- KU4.** Effective Communication Skills in Ethical Discussions
- KU5.** Cross-Functional Collaboration for Diverse Perspectives



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- KU6.** Advanced Deep Learning Models for Ethical Considerations
- KU7.** Ethical Education and Training Strategies
- KU8.** Importance of Ethical Principles in AI Projects
- KU9.** Development of Educational Materials in AI Ethics
- KU10.** Thought-Provoking Content Creation for AI Ethics
- KU11.** Training Program Design for Ethical Awareness
- KU12.** Mentoring and Knowledge Transfer
- KU13.** Latest Research and Guidelines in AI Ethics
- KU14.** Guidance on Ethical Implications in AI Decisions
- KU15.** Leading Cross-Functional Teams in Ethical Awareness
- KU16.** Frontline Knowledge in AI Ethics and Data Science



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Provide Educational Outreach and Ethical Guidance</i>	30	40	10	20
PC1. Utilise visual aids, storytelling, and practical examples to enhance the understanding of AI ethical concepts	-	-	-	-
PC2. Address ethical concerns transparently and foster open dialogue with stakeholders	-	-	-	-
PC3. Educate team members about ethical AI practises, fostering a culture of responsibility and ethical consciousness within the organisation	-	-	-	-
PC4. Communicate the importance of ethical principles in AI projects and applications	-	-	-	-
PC5. Develop training and educational materials on AI ethics for internal and external audiences	-	-	-	-
PC6. Develop thought-provoking content, such as articles and presentations, that bridges the gap between Data Science and Responsible AI	-	-	-	-
PC7. Conduct training sessions and programs to raise awareness among AI developers and users about ethical considerations	-	-	-	-
PC8. Mentor junior consultants, drawing from Data Science experience to nurture their growth	-	-	-	-
PC9. Stay informed about the latest research and guidelines in AI ethics to align policies with evolving standards	-	-	-	-
PC10. Provide timely advice on the ethical implications of AI decisions to guide decision-makers	-	-	-	-
PC11. Lead cross-functional teams, comprising Data Scientists, AI engineers, designers, and other experts, to cultivate a culture of ethical awareness, guidance, and responsible AI practices	-	-	-	-
NOS Total	30	40	10	20



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National Occupational Standards (NOS) Parameters

NOS Code	MSU/ITS/N1001
NOS Name	Facilitate Ethical Communication and Educational Outreach with Stakeholders
Sector	Information Technology Sector
Sub-Sector	
Occupation	AI DEVELOPMENT
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2028
NSQC Clearance Date	01/11/2023



Qualification Pack

MSU/ITS/N2001: Collaborate with Cross-Functional Teams for Ethical AI Development

Description

This OS unit is about facilitating ethical discussions, ethical integration and transparency in AI development, and applying ethical challenges to real-world scenario

Scope

The scope covers the following :

- Facilitate Ethical Discussions in AI Development
- Facilitate Ethical Integration and Transparency in AI Development
- Apply Ethical Challenges to Real-World Scenario

Elements and Performance Criteria

Facilitate Ethical Discussions in AI Development

To be competent, the user/individual on the job must be able to:

- PC1.** Facilitate productive discussions to address ethical considerations throughout AI development
- PC2.** Promote a culture of open dialogue and respectful exchange of ideas among cross-functional teams
- PC3.** Lead cross-functional workshops to brainstorm and develop ethical solutions for AI challenges
- PC4.** Collaborate with experts from various fields to explore innovative approaches to ethical AI development

Facilitate Ethical Integration and Transparency in AI Development

To be competent, the user/individual on the job must be able to:

- PC5.** Develop mechanisms to integrate transparency and ethical considerations into AI decision-making processes
- PC6.** Ensure that ethical considerations are transparently communicated in AI-related decisions
- PC7.** Collaborate with AI engineers to implement transparency features, ensuring ethical aspects are visible to stakeholders
- PC8.** Establish mechanisms to track and document ethical decisions made throughout the AI development and deployment processes
- PC9.** Regularly assess AI systems for potential biases and ethical challenges, contributing to continuous improvement
- PC10.** Balance technological advancements with ethical responsibilities in critical AI decisions
- PC11.** Track record in assessments such as Fairness/Bias, Explainability, Model Validation and Robustness, to assess model behavior through a Responsibility lens
- PC12.** Consider the potential social, political, environmental, and fairness implications of AI advancements

Apply ethical challenges to real-world scenario



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To be competent, the user/individual on the job must be able to:

- PC13.** Study case scenarios across diverse AI applications and industries
- PC14.** Identify common ethical challenges and unique considerations in various domains
- PC15.** Analyse and apply ethical principles to complex real-world AI ethics dilemmas
- PC16.** Develop ethical solutions that consider a wide range of factors, including societal impact
- PC17.** Collaborate to develop strategies for addressing the unique ethical challenges posed by emerging technologies
- PC18.** Engage in ethical hackathons to solve intricate AI problems with cross-functional teams

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Stakeholder Feedback Mechanisms
- KU2.** Ethical Adaptability in AI Landscape
- KU3.** Ethical Consideration Facilitation
- KU4.** Cross-Functional Workshop Facilitation
- KU5.** Innovative Approaches to Ethical AI Development
- KU6.** Transparency Integration in AI Decision-Making
- KU7.** Ethical Communication in AI-Related Decisions
- KU8.** AI Engineer Collaboration for Ethical Transparency
- KU9.** Ethical Decision Tracking and Documentation
- KU10.** Bias and Ethical Challenge Assessment
- KU11.** Model Behavior Assessment through Responsibility Lens
- KU12.** Societal, Political, Environmental, and Fairness Implications in AI
- KU13.** Ethical Challenge Identification in Various Domains
- KU14.** Application of Ethical Principles to AI Dilemmas
- KU15.** Holistic Ethical Solution Development
- KU16.** Unique Ethical Challenges in Emerging Technologies
- KU17.** Ethical Problem Solving in Cross-Functional Teams
- KU18.** Application of Data Science in Real-Life Business Problem Solving
- KU19.** Real-World Business Challenge Application of Ethical AI



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Facilitate Ethical Discussions in AI Development</i>	7	10	2	5
PC1. Facilitate productive discussions to address ethical considerations throughout AI development	-	-	-	-
PC2. Promote a culture of open dialogue and respectful exchange of ideas among cross-functional teams	-	-	-	-
PC3. Lead cross-functional workshops to brainstorm and develop ethical solutions for AI challenges	-	-	-	-
PC4. Collaborate with experts from various fields to explore innovative approaches to ethical AI development	-	-	-	-
<i>Facilitate Ethical Integration and Transparency in AI Development</i>	15	20	4	10
PC5. Develop mechanisms to integrate transparency and ethical considerations into AI decision-making processes	-	-	-	-
PC6. Ensure that ethical considerations are transparently communicated in AI-related decisions	-	-	-	-
PC7. Collaborate with AI engineers to implement transparency features, ensuring ethical aspects are visible to stakeholders	-	-	-	-
PC8. Establish mechanisms to track and document ethical decisions made throughout the AI development and deployment processes	-	-	-	-
PC9. Regularly assess AI systems for potential biases and ethical challenges, contributing to continuous improvement	-	-	-	-
PC10. Balance technological advancements with ethical responsibilities in critical AI decisions	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Track record in assessments such as Fairness/Bias, Explainability, Model Validation and Robustness, to assess model behavior through a Responsibility lens	-	-	-	-
PC12. Consider the potential social, political, environmental, and fairness implications of AI advancements	-	-	-	-
<i>Apply ethical challenges to real-world scenario</i>	8	10	4	5
PC13. Study case scenarios across diverse AI applications and industries	-	-	-	-
PC14. Identify common ethical challenges and unique considerations in various domains	-	-	-	-
PC15. Analyse and apply ethical principles to complex real-world AI ethics dilemmas	-	-	-	-
PC16. Develop ethical solutions that consider a wide range of factors, including societal impact	-	-	-	-
PC17. Collaborate to develop strategies for addressing the unique ethical challenges posed by emerging technologies	-	-	-	-
PC18. Engage in ethical hackathons to solve intricate AI problems with cross-functional teams	-	-	-	-
NOS Total	30	40	10	20



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National Occupational Standards (NOS) Parameters

NOS Code	MSU/ITS/N2001
NOS Name	Collaborate with Cross-Functional Teams for Ethical AI Development
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence
NSQF Level	6
Credits	7
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2028
NSQC Clearance Date	01/11/2023



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DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping Madhavi Skills University (MSU)
- Assessment agency deploys the ToA-certified Assessor for executing the assessment
- MSU monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time for the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SMEs
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from the assessment location
- Centre photographs with signboards and scheme-specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard and copies of the documents are stored

On the Job:

1. Each module (which covers the job profile) will be assessed separately.
2. The candidate must score 60% in each module to complete the OJT.
3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and aptitude are:
 - Videos of Trainees during OJT
 - Portfolio of items produced.
 - Viva
 - Demonstrations
4. Assessment of each Module will ensure that the candidate is able to:
 - Perform the skills mentioned in the assessment criteria.
 - Apply underpinning knowledge and understanding to the tasks performed to achieve desired level of competencies
 - Demonstrate behavioral and attitudinal soft skills as mentioned in the assessment criteria.



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Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSU/ITS/N2003. Identify and Assess the Ethical Risks of AI Systems	30	40	10	20	100	20
MSU/ITS/N2002. Develop and Implement Ethical Guidelines and Policies for AI Development and Use	30	40	10	20	100	30
MSU/ITS/N1001. Facilitate Ethical Communication and Educational Outreach with Stakeholders	30	40	10	20	100	20
MSU/ITS/N2001. Collaborate with Cross-Functional Teams for Ethical AI Development	30	40	10	20	100	20
DGT/VSQ/N0103. Employability Skills (90 Hours)	20	30	-	-	50	10
Total	140	190	40	80	450	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.